1. Data process

- How do you process the data?
  - Resize to adapt to model input size,

efficientnet	b0	b5	b6	b7
size	(224 × 224)	(456 × 456)	(528 × 528)	(600 × 600)

  - Data augmentation for
    - RandomHorizontalFlip(), [default p=0.5]
    - RandomRotation(10),
    - Normalize for 3 (mean, std) combination
      1. (0.485, 0.229)
      2. (0.456, 0.224)
      3. (0.406, 0.225)

2. Model architecture

- What is your model architecture?
  - efficientnet_b0 (4.01M), b5 (28.34M), b6 (40.74M), b7 (63.79M)
  - Single model b6 perform best

    2_b6.csv
    Complete · 3d ago
    0.99968

  - Ensemble perform same

    ensemble_b5_b5_pseudo_b6.csv
    Complete · 5m ago
    0.99968

- Do you make any changes/modifications to the model? Does these modifications improve performance?

  No modification.

3. Model training

- How do you train the model?

  Get the best valid accuracy as the result model

  - train : valid = 99 : 1
  - n_epoch all 10
  - lr all 1e-4, weight_decay all 1e-5

```
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=lr, weight_decay=weight_decay)
scheduler = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=n_epochs)
```

efficientnet	b0	b5	b6	b7
batch_size	32	8	8	2

4. Special techniques to improve performance

- Pseudo label

When epoch = 4~6, keep rejudge all unlabeled, if threshold > 0.9, “current epoch” put them into training set. Seems that result not become better.

5_b5_pseudo.csv Complete · 14s ago	0.99921
1_b5.csv Complete · 6d ago	0.99937

- Ensemble

Use b5, b5_pseudo, b6 to make majority vote (28.34+28.34+40.74 = 97.42 M)

ensemble_b5_b5_pseudo_b6.csv Complete · 5m ago	0.99968

1_b5.csv Complete · 6d ago	0.99937
5_b5_pseudo.csv Complete · 14s ago	0.99921
2_b6.csv Complete · 4d ago	0.99968

Seems that result not always become better.

5. Other details want to mention (that improve the performance)

Although this HW can’t know on the leaderborad, but below methods may improve performance. (Only think, no implement.)

- TTA (Test-Time Augmentation)

  Due to training time we do the data augmentation, maybe do TTA then majority vote may make the prediction result more robust.

Appendix page - Kaggle public leaderboard

- Best single model